

package format, so that the bundle of files and libraries can be created and distributed.

Working with Behat and the syntax structure

In the basic format, the scenarios are written in natural language. A few descriptions of the various scenarios are given below.

- *Given some context*
- *And more context*
- *When some event*
- *And second event occurs*
- *Then outcome*
- *And another outcome*
- *But another outcome*

There's no actual difference between *Then*, *And*, *But* or any of the other words that start each line. These keywords are all made available so that your scenarios are natural and readable.

To work with different examples of BDD, the following scenarios can be analysed.

Example 1 — Internet search: If a user is willing to search the tutorials on 'Blockchain Programming' on the Internet, then follow the sequence below:

1. The Internet should be connected
2. Open the Web browser
3. Open a search engine
4. Enter the search string or keyword 'Blockchain Programming'
5. View results

This process can be followed using Behat as shown below.

Feature: Blockchain Programming

In order to see tutorials

As a researcher

I need to be able to get the tutorials and programming source codes of Blockchain Programming

The above example is a very basic scenario of using the Behat framework for BDD programming, in which the feature and its required aspects are specified.

Example 2 -- Online search and shopping from an e-commerce portal: In this example, the scenario is to search for a suitable smartphone from an e-commerce portal. There are certain conditions and rules that need to be met.

Feature: Smartphone Search and Online Order

In order to buy mobile phone

As a customer

I need to be able to search and order online for the smartphone with certain conditions

The rules are:

- Tax is 10 per cent
- Courier delivery charge for the shopping cart under INR 20,000 is INR 100

- Courier delivery charge for shopping cart over INR 20,000 is INR 200

Scenario: Buying a single smartphone under INR 20,000

Given there is a mobile cover that costs INR 100

When I add the 'Mobile cover' to the shopping cart

Then I should have 1 product in the shopping cart

And the overall price of what's in the shopping cart should be under INR 23,000

Scenario: Buying a single smartphone over INR 20,000

Given there is a 'Mobile cover' that costs INR 100

And I add the 'Power bank', which costs INR 5000

When I add the 'Mobile cover' to the shopping cart

And I add the 'Power bank' that costs INR 5000

Then I should have 1 product in the shopping cart

And the overall price of what's in the shopping cart should be under INR 30,000

Once the instructions are written, these scenarios are executed using the following command inside the project directory:

```
$ /bin/behat
```

On running this, the source code will be associated and tested with the test cases, and the test scenarios will be displayed as 'failed' or 'passed'.

For example, if *amazon.in* is the website mentioned in the source code for analysis and test automation, this website will be scanned with the mentioned conditions of price, shopping cart and other parameters in the scenario, so that the user can get the accurate output. This approach gives the software quality assurance team a detailed view of the outputs that can be generated using different phrases and idioms mentioned in the scenarios of the test project.

Scope for research and development

A number of FOSS test automation libraries and frameworks are available, with their own features and set of testing strategies. These tools can be used for the scenarios discussed in the article, and the results obtained will reveal how well they fare against Behat. **END** 

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